

CO4 AT 3-COMPARATIVE ANALYSIS

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1. Given a graph, implement Floyd's Algorithm to compute all-pairs shortest paths and Warshall's Algorithm to determine the transitive closure of the graph. Compare their objectives, outputs, dynamic programming approach, time complexity, and scalability.

CODE:

```
def warshall(graph):
    n = len(graph)
    reach = [[1 if graph[i][j] != INF else 0
              for j in range(n)] for i in range(n)]

    for i in range(n):
        reach[i][i] = 1

    for k in range(n):
        for i in range(n):
            for j in range(n):
                reach[i][j] = reach[i][j] or (
                    reach[i][k] and reach[k][j]
                )

    return reach

# Sample weighted graph
graph = [
    [0, 5, INF, 10],
    [INF, 0, 3, INF],
    [INF, INF, 0, 1],
    [4, 8, INF, 0]
```

OUTPUT:

Output

All-Pairs Shortest Paths:

[0, 5, 8, 9]

[99999, 0, 3, 4]

[99999, 99999, 0, 1]

[99999, 99999, 99999, 0]

Transitive Closure:

[1, 1, 1, 1]

[0, 1, 1, 1]

[0, 0, 1, 1]

[0, 0, 0, 1]

=== Code Execution Successful ===